



S. G. G. S. Institute of Engineering and Technology

Vishnupuri, Nanded-431606, India

Department of Civil Engineering

END SEMESTER EXAMINATION – 2025-26 (Semester-I)

Date: Monday, 01 December, 2025
Class: Third Year

Course: Environmental Engineering
Course Code: PCC-CE302

Time: 10:00 – 12:00
Maximum Marks: 60

- Note
1. Attempt all questions.
 2. Use of non-programmable calculator is permitted.
 3. Draw appropriate figures wherever required.

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Que. No.	Question statement				Mark	CO	BT														
1	Which Indian standard code laid down the water quality desirable and permissible limits for drinking water. What is the acceptable and permissible limits for pH and alkalinity in drinking water as per IS code.				06	CO1	BT1														
2	Explain working of slow sand filter with neat figure.				06	CO2	BT2														
3	Write a note on oxidation ponds system.				06	CO4	BT3														
4	The average monthly alum dose is 10 mg/L in a 20 MLD water treatment plant. Alum under sufficient alkalinity emanates carbon dioxide as presented in the reaction. How much quantity of alum shall be ordered to keep stock available for a month. Explain your decision on the quantity of alum to be procured (purchased). $Al_2(SO_4)_3 \cdot 18H_2O + Ca(HCO_3)_2 \rightarrow 2Al(OH)_3 \downarrow + 3CaSO_4 + 18H_2O + 6CO_2 \uparrow$				06	CO3	BT4														
5	Enumerate the technologies available for disposal of solid waste. Explain sanitary landfill with neat sketch.				08	CO6	BT3														
6	Forecast population using incremental increase method for a town having population as specified below for supplying water for 15 years. The plant commences its operation in year 2025. <table border="1"><tr><td>Year</td><td>1971</td><td>1981</td><td>1991</td><td>2001</td><td>2011</td><td>2021</td></tr><tr><td>Population</td><td>7000</td><td>15000</td><td>28000</td><td>45000</td><td>63000</td><td>90000</td></tr></table>				Year	1971	1981	1991	2001	2011	2021	Population	7000	15000	28000	45000	63000	90000	08	CO1	BT4
Year	1971	1981	1991	2001	2011	2021															
Population	7000	15000	28000	45000	63000	90000															
7	Design a standard rate trickling filter for a town with population of 50 thousand with water supply rate of 225 LPCD. The average BOD of domestic wastewater observed as 270 mg/L and about 25% BOD is removed in primary treatment. The effluent BOD shall not be more than 18 mg/L. Use NRC equation.				10	CO4, CO5	BT5														
8	A river due to excess sewage discharge turned into nala. The BOD ₅ measured at some point was 180 mg/L. It was observed that the rate of deoxygenation was 0.17/day (base e) at 20°C. The river flowing through the locality at average temperature of 35°C. How many days will it take to reduce BOD of river (nala) water to 25 mg/L.				10	CO4, CO5	BT3, BT4														



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END SEMESTER EXAMINATION – 2025-26 (Semester-I)

Date: Monday, 01 December, 2025
Class: Third Year (Common)

Course: Bridge Engineering
Course Code: OE-IS-CE301

Time: 10:00 – 12:00
Maximum Marks: 60

- Note
1. Attempt all questions.
 2. Use of non-programmable calculator is permitted.
 3. Draw appropriate figures wherever required.

Que. No.	Question statement	Mark	CO	BT																		
1	Define, 1. Pier 2. Linear Waterway 3. Free Board 4. Design discharge 5. HFL	10	CO3	BT1																		
2	Enlist different NDT tests. Explain any one of NDT test.	05	CO6	BT2																		
3	Explain with neat diagram on IRC class AA tracked loading on bridges.	05	CO4	BT2																		
4	Discuss the interaction between old concrete and new concrete.	05	CO6	BT2																		
5	Enlist the different types of bearings. Explain functions of bearing.	05	CO3	BT5																		
6	Explain with the neat diagram masonry arch bridge.	05	CO1	BT2																		
7	Enlist methods of erection of bridge superstructure. Explain erection of bridge by launching single girder.	05	CO5	BT2																		
8	A multi-span bridge cost estimate prepared based on data obtained through the field conditions and subsequent analysis presented in table below. Suggest the span for this multi-span bridge. Mention the reason behind your decision on provision of span. <table><tr><td>Span (m)</td><td>4</td><td>8</td><td>10</td><td>14</td><td>16</td></tr><tr><td>Super-structure cost (Rs.)</td><td>250000</td><td>270000</td><td>285000</td><td>300500</td><td>394000</td></tr><tr><td>Sub-structure cost (Rs.)</td><td>350150</td><td>360505</td><td>300000</td><td>286800</td><td>295466</td></tr></table>	Span (m)	4	8	10	14	16	Super-structure cost (Rs.)	250000	270000	285000	300500	394000	Sub-structure cost (Rs.)	350150	360505	300000	286800	295466	06	CO5	BT4
Span (m)	4	8	10	14	16																	
Super-structure cost (Rs.)	250000	270000	285000	300500	394000																	
Sub-structure cost (Rs.)	350150	360505	300000	286800	295466																	
9	Write the Identification codes for the following bridge conditions, 1. A fourth bridge at chainage of 15.600 m. 2. A seventh bridge between chainage 142 and 143 km on right side service lane.	06	CO1	BT3																		
10	Suggest a foundation depth based on hydraulic calculation for the river bridge. The flow of river is 165 m ³ /s. The waterway to be provided is 65 m. Assume Ksf as	08	CO2	BT4																		

	1.36. Consider the strata at river is 1.0 m of soft soil, followed by soft rock of 0.7 m and then hard rock. Explain the reason behind your decision on provision of foundation depth.			
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Course Outcome (CO)	Bloom's Taxonomy Level (BT)
CO1 Know the bridges and their classifications.	BT1 – Remember
CO2 Comprehend bridge hydraulic parameters.	BT2 – Understand
CO3 Know the components of bridges.	BT3 – Apply
CO4 Interpret the loadings on bridges.	BT4 – Analyse
CO5 Explain the construction steps for bridges.	BT5 – Evaluate
CO6 Articulate the testing and strengthening of bridges.	BT6 – Create

*** All the Very Best ***